THE COPPERBELT UNIVERSITY

SCHOOL OF MATHS AND NATURAL SCIENCES –

DIT 460/ BIT 275 TEST 2 22/10/2013

TIME: 1 hr: 30 minutes.

Q 1 Provide examples of insertion, deletion, and modification anomalies and hence or otherwise Describe and illustrate the process of normalizing the table shown in Figure below upto 3NF. **[10 marks]**

| EmpNo | ContractNo | HourPerweek | eName | hotelNo | HotelLocatio |
|---|---|---|---|---|---|
| 120092 | C23 | 18 | Mulenga Mutale | H25 | Lusaka |
| 129932 | C23 | 24 | Jane Phiri | H25 | Lusaka |
| 129834 | C24 | 28 | Andrew Mutafela | H20 | Kitwe |
| 120093 | C24 | 16 | Brian Nzala | H20 | Kitwe |

Q 2 Given the schema below, write relational algebra statements for the queries below. **[8 marks]**

Person (person_name, Age, salary, cellno); Company (company_name, managerName)
Works (person name, company name); Lives (person name, address, city);
Located_in (company name, city);

i) List the names of the persons who work for 'ZESCO' along with the cities they live in.

ii) Find the persons who work for 'ZESCO' with a salary of more than K10000. List the names of these persons along with the address and cities where they live.

iii) Find the names of the persons who live and work in the same city.

iv) Find the names of the persons who do not work for company 'ZESCO'.

Q 3 Using the relational schema in Q 2 above write SQL statements to do the following.

i) Create tables, Person, Company, Works, Lives and Located_in. **[ 4 marks]**

ii) Insert a tuple in the Person table supplying data for the Person_name and Salary. **[2 marks]**

iii) Update the salaries of persons whose lives in Lusaka by a reduction of 15%. **[2 marks]**

iv) Delete from the persons table all persons that have reached a retirement age of 65. **[2 marks].**

Q 4 Given the relational schema in Q 2, write SQL statements for the queries below. **[12 marks]**

i) Get the details of Persons with salaries above the average salary

ii) Get the details of companies that have more than 10 persons working in them.

iii) Find the total salary for persons working in companies located in Lusaka.

iv) Get the details of companies that are located in Lusaka and have less than 10 persons working in them.